

admissions_eda.ipynb

DePaul Admissions Intelligence — Exploratory Data Analysis Notebook

Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Load Dataset

```
# Load dataset

df = pd.read_csv('DePaul_Data.csv')

# Display first 5 rows
print(df.head())
```

Dataset Overview

```
# Dataset shape
print('Dataset Shape:', df.shape)

# Dataset information
print(df.info())

# Statistical summary
print(df.describe())
```

Missing Values Analysis

```
# Check missing values
missing_values = df.isnull().sum()

# Sort missing values descending
missing_values = missing_values.sort_values(ascending=False)
```

```
print(missing_values)
```

Duplicate Record Analysis

```
# Total duplicate rows
duplicate_rows = df.duplicated().sum()

print('Duplicate Rows:', duplicate_rows)

# Unique student IDs
unique_students = df['Reference_ID'].nunique()

print('Unique Students:', unique_students)
```

Country Distribution Analysis

```
# Top 10 countries by application count
country_counts = df['Country'].value_counts().head(10)

print(country_counts)
```

Visualization

```
plt.figure(figsize=(10,6))
country_counts.plot(kind='bar')

plt.title('Top 10 Countries by Applications')
plt.xlabel('Country')
plt.ylabel('Applications')
plt.xticks(rotation=45)
plt.tight_layout()

plt.savefig('country_distribution.png')
plt.show()
```

Programme Demand Analysis

```
programme_counts = df['Programme'].value_counts().head(10)

print(programme_counts)
```

Visualization

```
plt.figure(figsize=(10,6))
programme_counts.plot(kind='bar')

plt.title('Top Programmes by Applications')
plt.xlabel('Programme')
plt.ylabel('Applications')
plt.xticks(rotation=45)
plt.tight_layout()

plt.savefig('programme_analysis.png')
plt.show()
```

Counsellor Workload Analysis

```
counsellor_counts = df['Caller_Name'].value_counts()

print(counsellor_counts)
```

Visualization

```
plt.figure(figsize=(8,5))
counsellor_counts.plot(kind='bar')

plt.title('Counsellor Workload Distribution')
plt.xlabel('Counsellor')
plt.ylabel('Number of Contacts')
plt.tight_layout()

plt.savefig('counsellor_workload.png')
plt.show()
```

Intake Semester Distribution

```
semester_counts = df['Semester'].value_counts()

print(semester_counts)
```

Visualization

```
plt.figure(figsize=(8,5))
semester_counts.plot(kind='pie', autopct='%1.1f%%')

plt.title('Intake Semester Distribution')
plt.ylabel('')
plt.tight_layout()

plt.savefig('semester_distribution.png')
plt.show()
```

Application Source Analysis

```
source_counts = df['Application_Source'].value_counts()

print(source_counts)
```

Visualization

```
plt.figure(figsize=(8,5))
source_counts.plot(kind='bar')

plt.title('Application Source Distribution')
plt.xlabel('Application Source')
plt.ylabel('Count')
plt.tight_layout()

plt.savefig('application_source.png')
plt.show()
```

Outreach Outcome Analysis

```
outcome_counts = df['Outcome_1'].value_counts().head(10)

print(outcome_counts)
```

Visualization

```
plt.figure(figsize=(10,6))
outcome_counts.plot(kind='bar')

plt.title('Top Outreach Outcomes')
plt.xlabel('Outcome')
plt.ylabel('Frequency')
plt.xticks(rotation=45)
plt.tight_layout()

plt.savefig('outreach_outcomes.png')
plt.show()
```

Save Cleaned Dataset

```
# Save cleaned dataset

df.to_csv('cleaned_DePaul_Data.csv', index=False)

print('Cleaned dataset exported successfully.')
```



Key Insights Generated

- India accounted for the majority of applications.
- STEM and analytics programmes showed the highest demand.
- Fall semester dominated intake distribution.
- Counsellor workload was highly imbalanced.
- Several important fields contained 100% missing values.
- Duplicate records were operationally intentional.



End of Notebook